Writer's Cube V0.5 — Scope Document

Internal name: *The Quill* ***Phase:*** *Founder-only personal prototype (Phase 1 of 4)* ***Target users:*** *One — the founder* ***Target outcome:*** *The founder uses the product to draft her first novel, with AI helping her track her own story* ***Target duration:*** *14–16 weeks of build + sustained daily use* ***Status:*** *Frozen on signature. Any change to this scope requires a written addendum.*

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1. Purpose of V0.5

V0.5 is the **minimum surface the founder needs to start drafting a novel — with AI tracking her story as she goes.** It exists to:

1. Prove the writing experience feels good enough to use every day
2. Prove the AI Story Bible agent actually helps a writer not lose the thread of her own story
3. Generate a real friction journal from a real user (the founder) writing real prose with real AI assistance
4. De-risk both the foundation layer (writing, persistence) *and* the AI integration layer before exposing them to outside writers in V0.7

What changed from earlier scoping (May 2026): The founder has chosen to deprioritize visual environments (the Vintage Writer's Desk) and prioritize **AI agents from day one**, because the AI is what helps her actually finish a novel. Brand identity and immersive environments move to V0.9 (closer to launch), giving the founder months of lived dogfooding before visual decisions are made.

V0.5 is **not** a product. It is a tool the founder builds for herself, knowing the rest of the world will get something polished six months later.

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2. In scope

2.1 Writing surface (must work flawlessly)

- Rich-text editor based on **TipTap** (ProseMirror) with basic formatting: bold, italic, underline, three header levels, ordered and unordered lists, blockquote
- Browser-native spell check (no custom dictionary in V0.5)
- Word count: per scene, per chapter, per project
- In-scene find (Ctrl/Cmd-F)
- Project-wide substring search
- Adjustable body font size (3 sizes: comfortable / large / huge)

- One serif body typeface (Georgia or Charter) and one sans for UI

2.2 Project structure

- One active project (no multi-project switching yet)
- Project → Chapters → Scenes hierarchy
- Side navigation pane with collapsible tree
- Rename chapters and scenes inline
- Drag-to-reorder chapters and scenes
- Click to jump to any scene

2.3 Persistence (the trust layer — non-negotiable quality bar)

- **Auto-save** every keystroke, debounced ~1 second, to Supabase
- A visible "saved at HH:MM" indicator the founder can glance at
- Manual "Save now" button (for psychological comfort)
- **Daily versioned snapshots** of each scene, retained 90 days
- **Google Drive backup**: link account on first use; project state mirrored as a folder of `.md` files (one per scene) with daily snapshots in a `versions/` subfolder, plus a `manifest.json` describing structure
- **Local download**: one-click "Download project as ZIP" producing the same folder structure
- Side-by-side compare of any scene against a previous snapshot

2.4 Founder-priority writing features

- **Highlight-to-tag**
  - Five preset tags: `Look up`, `Revise`, `Weak`, `Fact check`, `Placeholder`
  - Highlight text → right-click → assign tag (or keyboard shortcut)
  - Tags are visually distinct in the document (subtle colored underline + icon in margin)
  - Dedicated "Tags" view in side nav: list of tagged passages grouped by tag type, each clickable to jump to its location
- **Typewriter mode**
  - Toggle from a single button (and keyboard shortcut)
  - Screen blacks out except for the current ~5 lines around the cursor
  - Spell check disabled
  - Editing of previously written lines gated by an "are you sure" confirm
  - Visible word count remains in a small corner element

2.5 AI integration (new in this scope revision)

V0.5 ships with two AI features, both routed through the founder's own Anthropic API key. The product proxies the call (so the key stays in Supabase secure storage), but the cost lands on the founder's Anthropic account.

- **Minimal BYO Key Setup**
  - A simple settings screen where the founder pastes her Anthropic API key
  - "Test key" button that runs a tiny request and confirms it works
  - Real-time cost-this-month indicator pulled from her Anthropic usage data (or estimated locally if Anthropic doesn't expose the data per-key)
  - *Note: this is the minimal version. The polished Guided AI Setup Wizard ships in V0.9 for outside users.*
- **Story Bible AI agent** *(the signature AI feature)*
  - Background agent that reads the project and maintains a structured "story bible":
    - Characters (name, description, traits, first appearance)
    - Places (name, description, scenes that occur there)
    - Timeline (chronological list of events with their scene-of-occurrence)
    - Relationships (character-to-character, with how the relationship has evolved)
  - Re-runs after each writing session (or on demand via a button) — uses prompt caching aggressively to keep cost low
  - Each entity is displayed in the navigation pane and is clickable to see source passages
  - Founder can ask the Story Bible questions in plain language: *"Did Sarah meet Marcus yet?" "What did the locket look like the last time it appeared?"*
  - This is the feature most likely to help her finish a novel — it removes the cognitive load of remembering her own story
- **Highlight + right-click + ask AI**
  - Highlight any passage, right-click, choose "Ask AI"
  - Opens a small inline panel where the founder can ask a question about the passage
  - AI has access to full project context (cached)
  - Use cases: "Is this scene's pacing too fast?", "What sensory details would I add here?", "Does this contradict anything earlier?"

Underlying model: Claude Sonnet 4.6 via Anthropic API. Model is configurable in settings but defaults to Sonnet 4.6. Prompt caching enabled by default.

Estimated founder API spend during V0.5: ~$5–15/month on her own Anthropic account, based on writing 3–5 days/week with Story Bible running at end of each session and ~10 highlight-ask queries per session. The brief's §13 cost table now includes this line.

2.6 Themes

- Light theme (default)
- Dark theme
- Both use sane, professional defaults. **No** Vintage Desk environment artwork or brand identity in V0.5 — those land in V0.9, after the founder has spent months drafting and can give the brand designer informed direction.

2.7 Authentication

- Single user (the founder) signs in via Google OAuth (re-uses the Drive integration)
- No signup flow, no marketing site, no terms acceptance UI

2.8 Hosting

- Deployed on a Vercel preview URL initially; optional `v05.writerscube.app` subdomain once the domain is purchased
- No custom marketing domain in V0.5

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3. Out of scope

These are deliberately deferred. Every one of them is somebody's "wouldn't it be nice if..." — but **adding any of them to V0.5 is a scope change that requires a written addendum.**

Feature	First scheduled phase
Branching narratives	V0.7
AI persona onboarding survey (voice/values setup)	V2.0
AI Research Assistant persona	V2.0
AI Thought Partner persona	V2.0
Character interview mode (chat *as* a character)	V2.0
Vintage Writer's Desk environment / any artwork-based environment	**V0.9 (decision-gated by §6 success criteria)**
Brand identity (logo, type, color)	**V0.9**
Futuristic Studio environment	V1.5
Customizable widgets	V1.5
Daily goals, streaks, pomodoro	V1.5
Project templates (by audience tier)	V0.9
Multiple-project support	V0.9
Sharing / beta-reader access	V0.9
Polished onboarding walkthrough + Guided AI Setup Wizard	V0.9
Subscription, payments, pricing UI	V0.9
Multi-user authentication & accounts	V0.9
Custom marketing domain	V0.9
Export to DOCX, PDF	V0.9 (Markdown export ships in V0.5)
TTS playback, audio	V2.5
Mobile / tablet UI	V3.0
Desktop wrapper (Tauri)	V3.0
Cube View (3D manuscript navigation)	V3.0

4. Technical stack

Layer	Choice	Rationale
Framework	Next.js 15+ (App Router)	Modern, well-supported, hosts cleanly on Vercel
Hosting	Vercel (Hobby tier)	Free, instant deploys
Database & auth	Supabase (Free tier)	Postgres + Google OAuth + secure secret storage for the API key
Editor	TipTap (ProseMirror)	Mature, extensible, has the primitives needed for V0.7's branching feature
Styling	Tailwind CSS	Fast iteration, no design system overhead yet
State	Zustand for client state; React Query for server state	Lightweight, low ceremony
Cloud backup	Google Drive API (scope: `drive.file`)	Limits access to files Writer's Cube creates
Local backup	Browser File System Access API + JSZip fallback	Native picker in Chrome/Edge; ZIP download as universal fallback
AI provider (in-product)	**Anthropic API (Claude Sonnet 4.6) — founder's BYO key**	**Strongest API privacy terms; prompt caching support; best long-context performance for the Story Bible agent**
AI orchestration	Anthropic SDK (TypeScript) called from Next.js server actions	Keeps the founder's API key server-side; no client-side key exposure
Development AI	Claude Max 20x via Claude Code	The build itself
Code hosting	GitHub (private repo under entity account)	Standard

Total V0.5 monthly running cost:

- $0 hosting (free tiers)
- $200 Claude Max 20x (development)
- ~$5–15 founder's Anthropic API spend (using the product)
- ~$1 domain (`writerscube.app`)
- **≈ $206–216/month**

5. Build timeline (14-week target)

Week	Milestone	Definition of done
1	Editor + project shell	Founder can open the app, create a project with chapters/scenes, type into the editor, and reload without losing content
2	Auto-save + side nav polish	Auto-save works reliably; "saved at HH:MM" indicator; drag-to-reorder; rename inline
3	Google Drive integration + ZIP export	OAuth flow works; project mirrored as `.md` folder; ZIP download produces a clean local archive
4	Daily versioned snapshots + side-by-side compare	Snapshots generated daily; compare-with-previous works
5	Highlight-to-tag system + Typewriter mode	All five tag types work; Tags view shows grouped list with jump-to; typewriter mode toggle, blackout, edit-confirm, restore-on-exit
6	Project-wide search + light/dark themes	Substring search; theme switcher
7	**Minimal BYO Key Setup**	Settings screen for the founder's Anthropic key; test-key button; key stored encrypted server-side
8	**Story Bible AI agent — extraction layer**	Background job extracts characters, places, timeline, relationships from project; results visible in nav pane
9	**Story Bible AI agent — query layer**	Founder can ask the Story Bible questions in plain language; receives accurate, citation-grounded answers
10	**Highlight + right-click + ask AI**	Inline panel; full-project-context queries; passes prompt caching test
11	AI cost tracking + prompt-cache optimization	Founder can see live monthly cost in the app; caching reduces per-call cost by ~50% versus uncached
12	Bug-squash + persistence-stress week	Founder writes daily for the full week; data integrity audited; AI accuracy spot-checked on her own manuscript
13–14	Sustained-use evaluation	Founder writes daily; weekly friction-journal review; only critical bug fixes shipped
15–16	Stretch buffer for slipped work and continued drafting	

By the end of week 16, V0.5 is closed. V0.7 planning begins.

6. Success criteria

V0.5 is successful if, by week 16:

1. **Founder has used the product on at least 3 days/week for 6 consecutive weeks**
2. **Zero data-loss incidents** — no scene ever lost more than the last 5 seconds of input
3. **Founder has drafted at least 15,000 words inside the product** (a meaningful chunk of a novel)
4. **Friction journal contains at least 40 entries** (the signal that drives V0.7 scope)
5. **Story Bible AI agent has been used at least 30 times and rated "useful" by the founder on ≥80% of those uses** (this validates the highest-ROI feature)
6. **Highlight-tag feature has been used at least 25 times**
7. **AI cost stays under $20/month** (validates the BYO economics for outside users)

If any of (1) through (3) fails, V0.7 does not begin until the failure is diagnosed and fixed in V0.5. If (5) fails (Story Bible isn't useful), the entire AI architecture is reconsidered before V0.7 ships AI to outside users.

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7. Anti-goals (what V0.5 explicitly is *not*)

- **Not a shippable product.** No marketing site. No signup flow. No payment integration. No terms of service UI.
- **Not screenshotable.** Visuals are functional, not branded. The Vintage Desk environment is V0.9. Logo is a wordmark in a default typeface.
- **Not multi-user.** No invitations, no sharing, no roles, no permissions.
- **Not optimized for large projects.** Performance is fine for a single 100k-word novel; it does not need to handle 10 novels at once.
- **Not feature-complete versus competitors.** Scrivener has dozens of features V0.5 doesn't. That's fine — V0.5's job is to be reliable, fast, trustworthy, and *intelligent* enough to help one user finish her novel.
- **Not the polished AI experience outside users will see.** The BYO key setup is minimal. The Guided AI Setup Wizard with personalized cost estimator ships in V0.9.

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8. Decision log

Date	Decision	Rationale
2026-05-11	Editor framework = TipTap	ProseMirror primitives needed for V0.7 branching; mature ecosystem
2026-05-11	Google Drive scope = `drive.file` only	Smallest possible permission ask; preserves user trust
2026-05-11	No multi-project in V0.5	One novel is the goal; schema can wait
2026-05-11	**Story Bible AI agent in V0.5** *(scope revision)*	Founder believes AI tracking is what will help her finish a novel — the highest-ROI feature for her as the first user
2026-05-11	**Vintage Desk environment moved to V0.9** *(scope revision)*	Visuals don't help the founder write; they help outside users be wowed. Deferred to closer to launch, with the benefit of informed direction from months of dogfooding
2026-05-11	Default AI model = Claude Sonnet 4.6	Best long-context + structured output performance for Story Bible; aggressive prompt caching cuts cost in half
2026-05-11	AI key stored server-side (encrypted), proxied via Next.js server actions	Never expose the key to client JS; key is bound to founder's account

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9. Sign-off

Scope frozen on: _____

Signed (founder): _____

Any change to this document requires a written addendum dated and signed below.

Addenda

(none yet)